

B1: Shape

B1: Correct labelled values (allow ecf)

- 3 (a)** uniform electric and magnetic fields normal to each other AND **B1**
charged particle enters region normal to both fields
correct B direction w.r.t. E for zero deflection **B1**
For no deflection, $v = E/B$ **B1**
- (b)** $r = mv / BQ$, so B proportional to m
 $B = (0.680) \times (87/86) = 0.6879 \text{ T}$ **C1**
 $\Delta B = 7.9 \times 10^{-3} \text{ T}$ **A1**